

12.4.1 Solutions

$$4. \frac{\binom{5}{2} + \binom{3}{2}}{\binom{8}{2}} = 0.4642857$$

$$5. 4/36 = 1/9 \quad 0.1111111$$

$$6. 4/16 = 1/4$$

$$7. \frac{6}{36} + \frac{5}{36} - \frac{1}{36}$$

$$8a. p(\text{large}) = \frac{1}{1000}; p(\text{small}) = \frac{6}{1000} - \frac{1}{1000} = \frac{5}{1000}$$

$$8b. p(\text{large}) = \frac{1}{1000}; p(\text{small}) = \frac{3}{1000} - \frac{1}{1000} = \frac{2}{1000}$$

$$8c. p(\text{large}) = \frac{1}{1000}; p(\text{small}) = \frac{0}{1000}$$

$$9. \frac{\binom{4}{1}\binom{13}{5}}{\binom{52}{5}} = 0$$

$$10. \frac{\binom{13}{1}\binom{4}{4}\binom{12}{1}\binom{4}{1}}{\binom{52}{5}} = 0.4705882$$

$$11. 1 - \frac{1}{2^8} = 0.0039063$$

$$12a. \frac{6}{6^5} = \frac{1}{6^4} = 7.7160494 \times 10^{-4}$$

$$12b. 1 - \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2}{6^5} = 0.9074074$$

$$13a. \frac{2}{7} = 0.2857143$$

$$13b. \frac{9}{49} = 0.1836735$$

$$13c. \frac{6}{36} = \frac{1}{6} = 0.1666667$$